The cancellation-aware error model, and the 23×80 model

Results on the 1,000 test spectra

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In brief

1. **The new error model works.** With a discrepancy $\varepsilon(m_{hh}) \sum_i |C_i a_i|$, fitted on the 38,076 unused training rows, the pulls on the test spectra have s.d. 1.004 and, unlike before, are calibrated at every degree of cancellation ($\langle z^2 \rangle$ = 0.97–1.05 in each fifth, against 0.90–1.32 with $a(m_{hh})\sigma_k^2$). The worst spectrum goes from $\langle z^2 \rangle = 12.4$ to 1.5.
2. **The 23×80 model is as good as the network.** A weighted linear fit of the 23 functions in each bin (1,840 numbers, no training) has the same accuracy, the same calibrated uncertainty with the new error model (s.d. 0.999), a slightly smaller parameter uncertainty (5.2% vs 6.1% of the noise), and nearly the same bin-to-bin smoothness. Its posterior is exactly Gaussian.
3. **A smaller network is not needed** to answer the question: the network's only advantage over the 23×80 model is smoothness, and that advantage is small.

1 The new error model

The predictive variance of bin k of a spectrum at θ is

$$s_{\text{tot}}^2 = s_k^2 + \varepsilon(m_{hh})^2 \Big(\sum_{i=1}^{23} |C_i(\theta)\, \bar{a}_i(m_{hh})| \Big)^2, \tag{1}$$

with s_k^2 the network (or fit) uncertainty and $\bar{a}_i$ the mean coefficient functions. ε^2 was determined bin by bin on the unused training rows by requiring $\langle z^2 \rangle = 1$, and smoothed with $\ln \varepsilon$ quadratic in m_{hh} (fig. 1, right). It rises from 0.06% at threshold to about 0.3% at 600 GeV and 4–5% at 1.4 TeV.

Overall both discrepancy forms give a pull s.d. of about 1.00, but only the new one is right for every kind of spectrum: the old one is too narrow where the terms cancel and too wide where they do not. The total width barely changes on average (median 1.02 times the noise, against 1.06); it moves from the ordinary spectra to the cancelling ones.

The rise of ε at high mass

At 1.4 TeV the discrepancy is 4–5% of the sum of the terms. Whether that reflects the precision of the generator at high mass, or something else (for instance the very small event counts there, about 20 per bin), should be part of the question to the colleague who produced the samples. It

model + discrepancy	s.d. of pull all	255–555	> 555	$\lvert z\rvert < 2$	$\langle z^2\rangle$ by cancellation fifth (1 = strongest … 5)
ensemble mixture, none	1.076	1.152	1.049	94.1%	1.52, 1.10, 1.08, 1.07, 1.03
ensemble mixture + $a(m_{hh})\sigma_k^2$	1.003	1.023	0.996	95.6%	1.32, 0.95, 0.94, 0.93, 0.90
ensemble mixture + $\varepsilon^2(\sum \lvert C_i a_i\rvert)^2$	1.004	1.013	1.000	95.5%	0.97, 0.98, 1.04, 1.05, 1.01
23×80, none	1.092	1.208	1.050	93.8%	1.66, 1.12, 1.09, 1.07, 1.03
$23 \times 80 + a(m_{hh})\sigma_k^2$	1.006	1.036	0.995	95.6%	1.39, 0.95, 0.93, 0.92, 0.88
$23 \times 80 + \varepsilon^2(\sum \lvert C_i a_i\rvert)^2$	0.999	1.009	0.996	95.5%	0.93, 0.97, 1.04, 1.05, 1.01

Table 1: Calibration on the 1,000 test spectra with the exact Poisson noise. Each discrepancy was fitted on the unused training rows; for a standard normal $\lvert z\rvert < 2$ holds for 95.4%.

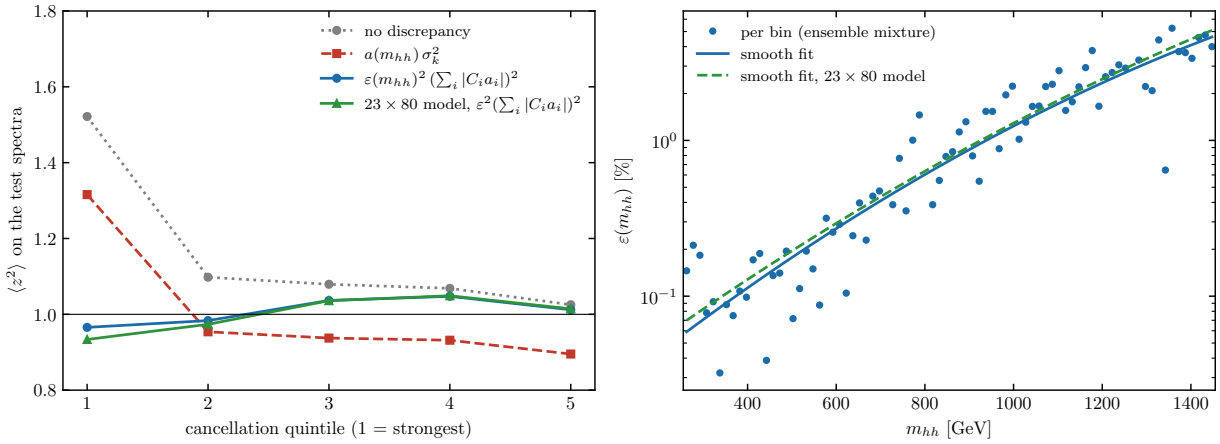


Figure 1: Left: $\langle z^2\rangle$ on the test spectra by degree of cancellation, for the two discrepancy models. Right: $\varepsilon(m_{hh})$ per bin and its smooth fit.

hardly matters for the cross section, which is small there, but it belongs in the paper.

2 The 23×80 model

In each of the 80 bins the cross section is linear in the 23 numbers $a_i(m_k)$, so with a Gaussian likelihood the weighted least-squares fit is also the exact posterior (for a flat prior): mean $\hat{\boldsymbol{a}}_k$, covariance $(X_k^{\mathsf{T}} W_k X_k)^{-1}$. No training, no random seeds, no approximation.

- **Accuracy**: median χ^2/N 1.088 (network ensemble 1.085); the same worst spectra.
- **Uncertainty**: the parameter uncertainty is 5.2% of the noise (median), against 6.1% for the Laplace mixture of the ensemble; with the new discrepancy, the pulls are calibrated as well as the network's (table 1).
- **Smoothness**: the rms second difference of the a_i across bins, relative to their size, is 0.070 for the 23×80 model and 0.059 for the network (median over the 23 functions).
- **Limitations**: values only at the 80 bin centres (enough for binned spectra); the poorly determined combinations (a_{16}–a_{20}) are as poorly determined as in the network, but their uncertainty

is exact and propagates correctly.

A natural refinement

If smoothness in m_{hh} is wanted, the principled way is a Gaussian-process prior on each $a_i(m_{hh})$ across bins: the model stays linear, the posterior stays exactly Gaussian, and the correlation length is set by the evidence. This is the "smooth 23×80 model", and it would answer the question about a smaller network more directly than training one.

3 Next steps

1. Decide which surrogate the paper presents: the network ensemble (as now), the 23×80 model, or both, as mutual cross-checks.
2. Adopt the cancellation-aware discrepancy in either case, and update the appendix, the figures and the executive summary.
3. Optionally, the smooth 23×80 model with a Gaussian-process prior.
4. The questions to the generator side, unchanged (`model_error_note.pdf`).